

Date: 08 JUL 2024

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Sub Code: 23CE/EE/ME/EC/CS/IT/AD/AM1T02

MIC23

B.Tech I Semester Supplementary Examinations, July-2024

LINEAR ALGEBRA AND CALCULUS

(Common to CIVIL, EEE, MECH, ECE, CSE, IT, AI&DS, AI&ML)

Time: 3 hours

Max Marks: 70M

- Note:** 1. Question paper consists of two parts (**Part A and Part B**)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART - A			10 X 2 = 20 M		
			Marks	BTL	CO
I.	a)	Find rank of the matrix $A = \begin{bmatrix} 8 & 1 & 3 \\ 6 & 0 & 3 \\ 2 & 2 & -8 \end{bmatrix}$	2M	L1	CO1
	b)	Test for the consistency of the system $x + y + z = 4, x - y + z = 0, 2x + y + z = 5$	2M	L2	CO1
	c)	Using Cayley-Hamilton theorem, find A^2 , given $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$	2M	L2	CO2
	d)	If 1,-2,3, are the Eigen values of the matrix of A, then find the Eigen values of $A^2 - 2A + 3I$	2M	L1	CO2
	e)	Define saddle point	2M	L1	CO3
	f)	State Rolle's mean value theorem	2M	L1	CO3
	g)	State Euler's theorem	2M	L1	CO4
	h)	Find $\frac{\partial u}{\partial x}$, where $u = x^2 \tan^{-1}\left(\frac{y}{x}\right)$	2M	L1	CO4
	i)	Evaluate $\int_0^1 \int_y^{\sqrt{y}} x^2 y^2 dx dy$	2M	L2	CO5
	j)	Evaluate $\int_0^1 \int_0^2 \int_0^3 (x + y + z) dx dy dz$	2M	L2	CO5
PART - B			5 X 10 = 50 M		
UNIT - I					
1.	a)	Reduce the following matrix into its normal form and hence find its rank $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$	10M	L3	CO1
(OR)					
2.	a)	Determine the values of α and β for which the system $x + 2y + 3z = 6, x + 3y + 5z = 9, 2x + 5y + \alpha z = \beta$ have (i). No solution (ii).Unique solution (iii). An infinite number of solutions	10M	L3	CO1
UNIT - II					
3.	a)	Reduce the quadratic form $3x^2 - 2y^2 - z^2 + 12yz + 8zx - 4xy$ to canonical form by orthogonal reduction and state the nature of the quadratic form	10M	L3	CO2
(OR)					

4.		Find the Eigen values and the corresponding Eigen vectors of $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$	10M	L2	CO2
UNIT – III					
5.	a)	Show that for , $x \in (0,1), x < -\log(1-x) < \frac{x}{(1-x)}$	5M	L2	CO3
	b)	Verify the Cauchy's mean value theorem for the following functions $f(x) = \sqrt{x}, g(x) = \frac{1}{\sqrt{x}}$ in $[a,b], 0 < a < b$	5M	L2	CO3
(OR)					
6.	a)	Calculate the approximate value of $\sqrt{10}$ to four decimal places using Taylor's theorem	5M	L3	CO3
	b)	Verify Rolle's theorem for the function $f(x) = x(x+3)e^{-x/2}$ in $(-3,0)$	5M	L2	CO3
UNIT – IV					
7.	a)	Find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $4x^2 + 4y^2 + 9z^2 = 36$	5M	L3	CO4
	b)	If $u = x+y+z, uv = y+z, uvw = z$, show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$	5M	L3	CO4
(OR)					
8.	a)	If $u = y^2 e^{y/x} + x^2 \tan(x/y)$ find the values of (i) $xu_x + yu_y$ and (ii) $x^2 u_{xx} + 2xyu_{xy} + y^2 u_{yy}$	5M	L3	CO4
	b)	Find the maximum and minimum values of the function $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$	5M	L2	CO4
UNIT – V					
9.	a)	Evaluate $\iint_R xy \, dx \, dy$, where R is the region bounded by X -axis, the line $x = 2a$ and the curve $x^2 = 4ay$	5M	L2	CO5
	b)	Evaluate $\int_0^a \int_y^a \frac{x}{x^2 + y^2} \, dx \, dy$ by changing the order of integration	5M	L2	CO5
(OR)					
10.	a)	Evaluate $\int_1^e \int_1^y \int_1^{e^x} \log z \, dx \, dy \, dz$	5M	L2	CO5
	b)	Find by triple integration, the volume of sphere $x^2 + y^2 + z^2 = 1$	5M	L2	CO5

*** End of the Question Paper ***